

Lecture 01

Health Informatics Introduction



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Assistant Professor Biomedical Engineering, PhD, UK

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Welcome to School of Medicine



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Who am I?

Nabeel Abdulrazzaq Fattah, PhD, Eng.

Educational background

PhD in Biomedical Engineering	UK
Diploma in Human Resource	Ireland
Diploma in Human Anatomy	Ireland
Diploma in Nursing and Patient Care	Ireland
Nursing Studies - The Physical Examination	Ireland
MSc in Nuclear Physics (1st honor)	IRAQ
BSc in General Physics (1st honor)	IRAQ

Work position

Visual Prosthesis Specialist, UK
College of Medicine / HMU / MHESR

Horses for courses

- What does this phrase mean?
- Give me an examples
- How do you connect with yourself as a medical student?





Medical student personality

Medical students often exhibit a variety of personalities and traits, shaped by the demands of their rigorous training. Here are some common characteristics:

- 1.Resilient:** They face high stress and long hours, so resilience is crucial for managing challenges and setbacks.
- 2.Empathetic:** A strong sense of empathy helps them connect with patients and understand their experiences.
- 3.Detail-oriented:** Attention to detail is vital for diagnosing and treating patients effectively.
- 4.Curious:** A natural curiosity drives their desire to learn and stay updated with medical advancements.

5. **Collaborative:** Teamwork is essential in healthcare, fostering collaboration with peers and other professionals.
6. **Motivated:** A strong intrinsic motivation to help others and pursue a medical career propels them through tough times.
7. **Organized:** Managing a heavy workload requires good organizational skills to balance studies, clinical rotations, and personal life.
8. **Ethical:** A strong commitment to ethics and professionalism guides their interactions with patients and colleagues.

Medical student etiquette in class

1. Punctuality

Arrive on time to class to show respect for the instructor and your peers.

2. Active Participation

Engage actively in discussions, ask questions, and contribute to group work without dominating the conversation.

3. Respectful Communication

Use appropriate language and tone when interacting with classmates and instructors. Be mindful of differing opinions.

4. Listening Skills

Listen attentively when others are speaking. Avoid interrupting, and wait for your turn to contribute.

5. Professional Appearance

Dress appropriately, especially during clinical skills sessions or presentations. Follow any dress codes set by the institution.

6. Use of Technology

Limit phone usage and social media during class. If laptops are allowed, use them for note-taking rather than browsing unrelated sites.

7. Collaboration and Support

Be supportive of your classmates. Offer help and encouragement, especially during stressful periods like exams.

8. Respect for Faculty

Treat instructors with respect, addressing them appropriately and being open to feedback.

9. Confidentiality

Respect patient confidentiality in discussions and assignments, as this is fundamental in the medical field.

10. Feedback and Reflection

Be open to constructive criticism and use it as an opportunity for growth. Reflect on your learning and interactions.

Which type of mobile phones are most popular among students?

1. Apple
2. Samsung
3. Xiaomi
4. OnePlus
5. Google
6. Realme
7. Huawei
8. Oppo
9. Motorola
10. Nokia

Problem

- A research problem is a specific issue, contradiction, or gap in knowledge that a researcher aims to address.
- A research problem serves as the foundation for the study, providing a clear focus and justification for the research.
- This problem can emerge from areas of concern, conditions needing improvement, or real-world challenges faced in a particular field.

Type of research

Descriptive Research:

Aims to describe characteristics of a population or phenomenon.

Analytical Research:

Involves the analysis of existing data to identify patterns or relationships.

Experimental Research:

Involves manipulating variables to determine cause-and-effect relationships.

Qualitative Research:

Focuses on understanding concepts, thoughts, or experiences through interviews, focus groups, and observations.

Quantitative Research:

Uses numerical data to quantify behaviors, opinions, or other defined variables.

Mixed-Methods Research:

Combines both qualitative and quantitative approaches to provide a comprehensive understanding of a research problem.

Longitudinal Research:

Studies the same subjects over a long period to observe changes over time.

Cross-Sectional Research:

Analyzes data from a population at a specific point in time.

Case Study Research:

In-depth examination of a single case or a small number of cases.

Action Research:

Focuses on solving a specific problem within a community or organization through a collaborative approach.

Practical Session



Survey

- a survey is a type of research method used to collect data from a specific population.
- It typically involves asking participants a series of questions to gather information on their opinions, behaviors, or characteristics.
- Surveys can be conducted in various formats, such as **online questionnaires**, **phone interviews**, or **in-person interviews**.

Example

1. What type of mobile phone do you currently use?

- ☐ iPhone
- ☐ Samsung
- ☐ Xiaomi
- ☐ OnePlus
- ☐ Other (please specify): _____

Example

2. How satisfied are you with your current mobile phone?

- ☐ Very satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very dissatisfied

Example

3. What features do you consider most important when choosing a mobile phone? (Select up to three)

- ☐ Camera quality
- ☐ Battery life
- ☐ Price
- ☐ Brand reputation
- ☐ Storage capacity
- ☐ Design and aesthetics
- ☐ Operating system (iOS/Android)

Example

4. How often do you upgrade your mobile phone?

- ☐ Every year
- ☐ Every 2 years
- ☐ Every 3-4 years
- ☐ I do not upgrade

Example

5. What is your primary reason for choosing your current mobile phone?

- ☐ Recommendations from friends/family
- ☐ Online reviews
- ☐ Price
- ☐ Features
- ☐ Brand loyalty

6. Would you be interested in participating in a follow-up interview about your mobile phone preferences?

- ☐ Yes
- ☐ No

Project plan (Survey)

- MS Word
- MS Excel
- MS PowerPoint



Project (Survey) - (Practical Session)

- Week 1 - Project title, writing 10 questionnaires
- Week 2 – Questionnaires form using Word
- Week 3 – 40 Participant, Entering data and data analysing using Excel
- Week 4 - Power point slides

Sources

- Google
- YouTube
- Gemini (Google AI)
- ChatGPT (AI)

Theoretical Session



Lecture Outline

- Week 1: Introduction to Windows + Microsoft Word
- Week 2: Microsoft Excel + Microsoft PowerPoint
- Week 3: Advanced Word Features + Advanced Excel Features
- Week 4: Advanced PowerPoint Features + Writing an academic email
- Week 5: Technology and Medicine + Artificial Intelligent and Medicine
- Week 6: Telemedicine + How to manage your clinical office

Thanks

